



## Department of Computer Science

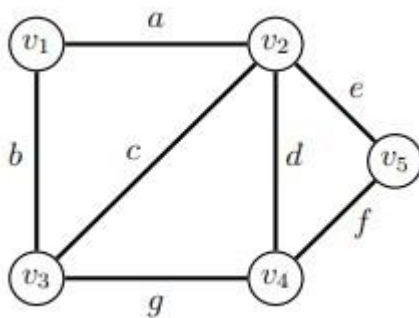
National University of Computer and Emerging Sciences  
Lahore Campus

### Data Mining

#### Section A

#### Quiz 3a

1) Find graph Laplacian of the following graph. (3 marks)



$$= D - A$$

$$D = 2, 0, 0, 0, 0$$

$$0, 4, 0, 0, 0$$

$$0, 0, 3, 0, 0$$

$$0, 0, 0, 3, 0$$

$$0, 0, 0, 0, 2$$

$$A = 0, 1, 1, 0, 1$$

$$1, 0, 1, 1, 1$$

$$1, 1, 0, 1, 0$$

$$0, 1, 1, 0, 1$$

$$0, 1, 0, 1, 0$$

Rest you can solve yourself by doing  $D - A$

.Some students may have written direct answer that is also acceptable.

**2) Is the calculated graph Laplacian rank deficient. If so, then why, show mathematical prove? (3 marks)**

**If they have shown that Since  $L1 = 0$  by multiplying L matrix with 1 vector, give them full marks.**

Use mathematical explanation to show whether this graph connected or not? **(3 marks)**

**If they have shown that or written that since second smallest eigen value of L is not equal to zero that is why graph is connected give them full marks.**

**4) Calculate graph density of above graph (1 mark)**

$$Ne/Nv^2$$

$$= 7/25 = 0.28$$

They may have used another formula but answer should be same.